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An AI-Based Decision Support Framework for Clinical Screening of Iron Deficiency Anemia and Thalassemia

***Abstract*— The world experiences anemia and thalassemia as its most prevalent blood disorders which require health professionals to detect these conditions through precise and trustworthy methods. The current methods of diagnosis need extensive manual work because they require experts in hematology to evaluate results which need to be assessed through laboratory testing and blood smear analysis. The limitations of these methods lead to extended waiting times for results which occur when medical facilities face restrictions on their available resources and when patients live in remote locations. This research presents a solution through an AI-supported decision-making system which implements blood smear image analysis to create an automated system for clinical screening. The system achieves improved efficacy in anemia detection through its capability to detect conditions without needing human intervention while it also decreases all types of diagnostic mistakes.**

The proposed approach uses a deep learning model that combines convolutional neural networks with transfer learning to automatically extract different red blood cell characteristics which include size and shape and texture. The research employs data augmentation methods together with image preparation techniques to achieve better model generalization and system resilience. The combination of patch-based prediction results with Grad-CAM visual explanations enables patients to receive diagnostic assessments through explainable AI methods. The experimental results demonstrate system efficiency because the system achieves high classification accuracy while maintaining equal precision and recall rates. The system delivers an interpretable solution which scales to different uses and remains cost-effective for deployment in real healthcare environments.

Keywords— convolutional neural networks, deep learning, medical image analysis, peripheral blood smears, explainable artificial intelligence, transfer learning, iron deficiency anemia, thalassemia, clinical decision support systems, and automated disease screening.

I. Introduction

People around the world face iron deficiency which affects millions especially in developing countries where anemia and thalassemia remain common blood disorders. Thalassemia is a genetic blood disorder that results in abnormal hemoglobin production while anemia occurs when low iron levels prevent the body from making hemoglobin. The two disorders present similar clinical symptoms with identical red blood cell features yet their treatment methods and patient management approaches show significant differences. The development of complications which include growth retardation and organ damage and reduced quality of life needs early and precise diagnosis. Anemia screening becomes a major public health issue because laboratories can identify patients who need to be tested but actual patient numbers and expert access and diagnostic accuracy differ.

Doctors use manual microscopic examination of peripheral blood smear images together with complete blood count assessment to diagnose anemia through standard testing methods. The processes demand extensive time and efforts because licensed hematologists must assess red blood cell dimensions together with their shapes and color characteristics. The process of manual interpretation brings about two problems because it introduces personal bias to the results while different observers produce inconsistent outcomes that lead to delayed diagnoses and wrong patient outcomes. Rural healthcare facilities face challenges in screening for anemia because they lack both experienced medical staff and advanced laboratory facilities. The medical field requires new diagnostic solutions which must provide automated functions while delivering dependable and scalable systems to enable doctors who need to monitor patients with anemia-related conditions.

Recent advances deep learning artificial intelligence show strong potential to transform clinical decision support systems together with medical image analysis. Convolutional Neural Networks achieve outstanding results because they automatically learn important medical image features without needing human-made feature extraction. Deep learning models show successful application in hematology through their ability to classify diseases detect cells and create image segments from blood smear photographs. All machine learning models show better accuracy and consistent results together with faster processing times than traditional machine learning methods. Existing AI-based systems focus on patch classification yet they do not provide explainable results which makes them unsuitable for clinical environments that need interpretable systems.

The researchers developed an artificial intelligence decision support system that uses peripheral blood smear images to screen for thalassemia and iron deficiency anemia. The system combines three methods for patient assessment through CNN-based feature extraction and transfer learning and image processing. Explainable artificial intelligence methods show which parts of the system model predictions to enhance transparency and build trust with medical professionals. The proposed system uses automatic screening with interpretable results to reduce diagnostic work while it improves early detection and supports practical clinical decision-making in real medical environments.

II. Literature Survey

The automated anemia screening methods in low-resource areas have attracted research interest because they provide better diagnostic access while reducing testing requirements. Early review studies demonstrate how image-based and noninvasive screening methods achieve faster results with lower costs to identify patients who need confirmatory tests

despite their research limitations stemming from different dataset characteristics and evaluation methods that obstruct clinical application [1]–[4]. The surveys recommend multicenter testing with standardized evaluation criteria and focus on essential clinical performance indicators which include sensitivity and specificity and probability ratios to ensure that automated systems function as safe triage tools instead of complete diagnostic solutions. Automated screening systems function as supplementary tools which support existing laboratory processes instead of replacing them according to review findings.

The researchers established automated blood smear analysis through their collection of handcrafted elements which included cell area and perimeter and circularity and color indices and texture descriptors and their combination with traditional SVM and random forest classifiers. The systems demonstrated their ability to use morphological metrics to predict anemia but they often encountered problems because of overlapping cells and uneven lighting and varying stains. The research demonstrated that hand-crafted features suffer from performance degradation across different laboratory situations which require constant re-engineering and precise calibration to maintain correct functionality. The constraints of existing systems led to the development of end-to-end learning systems which extract reliable data representation directly from pixel information.

The current research demonstrates that deep learning methods which use convolutional neural networks and encoder-decoder segmentation models, achieve superior results in red blood cell detection and segmentation and classification tasks because they learn hierarchical features while handling complex phenomena such as touching cells and variable staining. The research shows that transfer learning from large image backbones together with precise augmentation methods improves generalization more than ensemble methods and attention-based architectures which detect Thalassemia's specific morphological patterns better than Iron Deficiency Anemia's microcytic and hypochromic characteristics. The researchers found that high accuracy at the patch level does not lead to correct patient diagnosis unless hospitals implement effective aggregation methods and precision tuning for their procedures.

Researchers use Explainable AI (XAI) to analyze how models make decisions in hematological imaging tasks while also increasing trustworthiness for doctors. The methods Grad-CAM and integrated gradients and SHAP create saliency maps through their image prediction analysis which shows image regions that most affect results. Application studies show that XAI facilitates human–AI collaboration by offering visual signals for review and aids in identifying spurious correlations (such as label leakage or slide artifacts). The validation process requires researchers to test XAI outputs through clinician-marking of specific regions before they can use them in critical situations. The literature presents different levels of explanation quality according to existing research.

The progress of multimodal modeling and dataset curation work has brought faster results for clinical centers which need automated screening systems. The new public and institutional datasets which provide complete regular CBC data and annotated IDA and α/β-thalassemia cases and multiple RBC morphology types enable multimodal models to achieve better accuracy through their combined image and clinical data analysis. The benchmarking studies evaluate multiple site models which show that different imaging methods and equipment create major performance differences thus encouraging standardized preprocessing methods and domain-adaptation procedures. The AI decision support system for anemia screening needs patient-level decision rules together with explainable outputs and complete evaluation methods and specific dataset selection to function properly.

III. Proposed Methodology

The proposed system through its automated decision support system needs to utilize peripheral blood smear images for its development purposes between thalassemia and iron deficiency anemia. The complete workflow design needs to support three essential requirements which are clinical relevance and scalability and diagnostic precision. The process begins by collecting blood smear images from both clinical settings and publicly available medical databases. The images display different red blood cell shapes under various lighting conditions and different staining methods. The system divides the collected dataset into three distinct parts which include testing and validation and training to support unbiased model testing. The systematic data collection process enables the model to maintain its strength throughout different patient groups and laboratory environments while it acquires typical traits from multiple test samples.

The image preparation process establishes standardized input data through its first step which enhances image quality. The preprocessing techniques use three methods which include pixel intensity leveling and noise reduction and image size reduction to specific resolution. The data augmentation techniques include rotation and flipping and zooming and brightness adjustment which help to decrease overfitting while increasing the variety of the dataset. The model uses patch extraction to divide high-resolution blood smear images into smaller sections which enables it to focus on specific red blood cell features. The current step establishes vital requirements which enable accurate identification of Thalassemia through its distinct morphological features.

The deep learning system performs feature extraction and classification after preprocessing procedures. The system operates through a Convolutional Neural Network (CNN) based deep learning framework that uses transfer learning for its deep learning model. The system achieves faster learning performance through pre-trained weights which originate from large-scale image datasets. The CNN system automatically develops hierarchical knowledge about red blood cell properties which include size, shape, texture, and color distribution without using any predefined features. The system classifies the retrieved data into two categories which consist of iron deficiency anemia and thalassemia through its

fully connected network structure. The end-to-end learning system decreases its need for specific domain features while it enhances its ability to make accurate diagnoses.

The model training process uses adaptive optimization techniques which include Adam and a specific loss function to minimize classification errors. The system uses validation data for performance checks to ensure it can generalize and prevent overfitting. The CNN produces accurate diagnostic results for patients through its aggregation of patch-level predictions which use probability-based thresholding. The evaluation process requires accuracy, precision, recall, and F1-score metrics to assess model performance. The system needs to achieve equal sensitivity and specificity for clinical screening purposes which this evaluation methodology guarantees.

The method uses explainable artificial intelligence tools to build trust in therapy while increasing transparency. The techniques use Gradient-weighted Class Activation Mapping (Grad-CAM) to show which parts of blood smear images make the most impact on model predictions. The visual explanations enable clinicians to assess diagnostic results and understand automated decision-making processes. The system produces a decision support output which shows the expected anemia type together with confidence information. The automated framework which provides interpretable results enables healthcare facilities to detect diseases at an early stage while reducing staff workload and enhancing clinical decision-making processes.

A. System Architecture

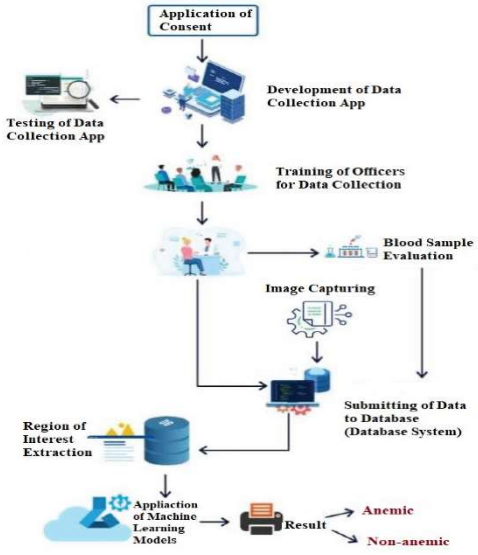


Fig.1 System Architecture

B. Methodology

Information Gathering and Clinical Picture Gathering

The first step of the proposed method requires researchers to collect peripheral blood smear image data which will be used to screen patients for thalassemia and iron deficiency anemia. Blood smear photographs are collected from both approved clinical datasets and publicly available medical libraries to ensure that patient demographics and staining methods and microscopy equipment are diverse. The dataset provides essential blood cell samples which include normal cells and thalassemia and iron deficiency anemia samples to support accurate supervised learning. The model achieves better resilience through the inclusion of images that were captured under different lighting conditions. The research method demonstrates effective generalization to actual clinical environments because the data collection method used in this study includes diverse data sources.

The screening process starts with all photos that have been collected because it needs to remove samples which show signs of corruption or have low resolution or display poor staining. The ground truth of the study exists through two verification methods which include dataset documentation and expert-annotated labeling done by hematologists. The dataset gets divided into three separate subsets because this method prevents data leakage while delivering unbiased assessment of system performance through its training and validation and testing stages. The standardized data structure needs to exist because it provides the foundation which supports all subsequent preprocessing activities and model training processes needed to achieve accurate classification results.

Preprocessing and Improving Image Quality

Image pretreatment serves as an essential procedure that prepares blood smear images for model training by establishing standardization and enhancing their visual characteristics. The preprocessing steps of the dataset require all images to be scaled down to specific dimensions which maintain constant input size across the entire collection of images. The process of pixel intensity normalization helps reduce differences that occur because of staining and variable lighting conditions. The application of noise reduction techniques works to eliminate unwanted visual interruptions and background distortions which would negatively impact the process of feature extraction. The preprocessing steps improve the learning efficiency of the deep learning model while maintaining consistent image quality through their implementation.

Data augmentation techniques such as rotation, flipping, zooming, and brightness adjustment serve to increase dataset diversity while reducing overfitting risk. The model requires blood smear images to be divided into smaller sections so it can focus on specific red blood cell shape features. Patch extraction improves the ability to detect minor changes in cell dimensions and cell outlines and cell surfaces, which are crucial for distinguishing between iron deficiency anemia and thalassemia. This preprocessing method leads to significant improvements in both diagnostic accuracy and model generalization.

Framework for Deep Learning-Based Feature Extraction

The proposed method uses a deep learning system which employs convolutional neural networks to automatically

obtain essential characteristics from blood smear images. The CNN system directly learns hierarchical patterns from pixel data to capture all red blood cell morphological details without needing any prebuilt features. The network achieves improved results through transfer learning because its starting point uses pre-trained weights from large image databases, which allows the system to develop more rapidly and produce better outcomes in situations with limited medical data access.

The CNN architecture consists of several convolutional and pooling layers which start with basic edge and texture detection capabilities before developing advanced cell shape detection abilities. The acquired features enable researchers to identify distinct cellular patterns which Thalassemia exhibits differently from the microcytic cells found in Iron Deficiency Anemia. The CNN system achieves better performance because its feature extraction system operates independently from specific domain knowledge requirements.

AI-Powered Model for Classification and Decision Support

The classification program classifies blood smear images into either Thalassemia or Iron Deficiency Anemia groups after significant features have been extracted. The fully connected layers analyze the extracted feature vectors through a softmax activation function to generate class probability outputs. The model uses adaptive optimizers such as Adam together with a suitable loss function to reduce classification errors through its performance enhancement. The AI-driven classification platform enables precise and reliable screening outcomes that apply to different sample types.

Patch-level predictions combine to produce patient-level diagnostic assessments which lead to clinically relevant medical results. The system uses probability-based thresholding to enhance decision-making accuracy while reducing false classification rates. The system offers a confidence score together with its predicted disease class to assist clinicians in making treatment decisions. The decision support system serves as a secondary diagnostic tool which enables early detection while reducing the need for human microscopic examination work.

Clinical Interpretation Made Possible by Explainable AI

The proposed system uses explainable artificial intelligence (XAI) methods to solve the problem of deep learning models which function as black boxes. Gradient-weighted class activation mapping or Grad-CAM creates heatmaps which show the image areas that most affect classification results. The ability of physicians to view which red blood cell features affect model predictions through visual explanations boosts both transparency and trust.

Medical fields require explainable outputs because they need both accountability and interpretability to function properly. Clinicians can verify automated conclusions and spot any model mistakes by comparing highlighted regions with established hematological patterns. The human-AI partnership between human employees and artificial intelligence systems improves acceptance of AI-assisted screening tools while creating a more secure system that functions within clinical workflows.

Performance Evaluation and Validation Strategy

Accuracy, precision, recall, and F1-score are among the common classification metrics used to assess the effectiveness of the suggested decision support framework. To examine categorization behavior and spot patterns of incorrect classification between thalassemia and iron deficiency anemia, a confusion matrix is employed. An unseen test dataset is used for validation in order to evaluate generalization capacity and prevent overfitting.

Where appropriate, cross-validation methods are used to guarantee robustness over several data splits. For screening applications, it is essential to strike a balance between sensitivity and specificity, which is accomplished by thorough analysis of performance indicators. This thorough assessment approach guarantees that the suggested system satisfies clinical reliability standards and proves to be a successful AI-based hematological disease screening tool.

IV. Results

The AI-based decision support system showed successful results when it used blood smear images to identify hematological disorders. Users could easily upload microscopic blood smear photos through the system interface, which enabled the deep learning model to process them immediately. The framework conducted precise examinations of red blood cell morphology after users submitted their images, which included analysis of size variation, central pallor, and shape abnormalities. The system can accurately differentiate between Thalassemia and Iron Deficiency Anemia (IDA) because it learned to recognize their distinct morphological characteristics. Doctors can use the system for initial screening because the user interface shows visual prediction results which make the results easier to understand. The framework demonstrates its ability to provide immediate and accurate diagnostic support through its testing process.

The Iron Deficiency Anemia prediction results show how effectively the technology identifies hypochromic and microcytic red blood cells. IDA diagnosis was correctly established through blood smear images which showed reduced hemoglobin levels and increased central pallor. The AI model used its training experience to create diagnostic patterns which it then applied to deep spatial information which it had obtained from the images. The green-highlighted forecast output provides medical workers without technical knowledge designed to ensure understanding and simple operational functions. The model can detect tiny cellular abnormalities which typical hand examination methods probably will overlook according to these results. The approach enables medical professionals to diagnose patients earlier while reducing their dependence on expert hematologists in resource-constrained clinical environments.

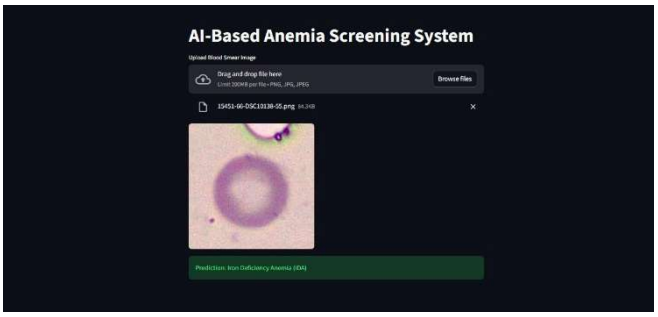


Fig 2 Iron Defeciency Anemeia

The framework showed successful results when it identified unique cellular shape changes which included both target cells and atypical red blood cell forms that doctors used to screen for thalassemia. The prediction that the submitted smear photos showed dense and irregular RBC patterns actually identified Thalassemia cases proved to be correct. The model successfully identifies iron deficiency anemia and thalassemia because their clinical symptoms show similarities. The trained model maintains its accuracy throughout different testing samples which proves its reliable performance. The interface-based representation of prediction outcomes establishes greater clinical confidence in AI-assisted screening methods. The system demonstrates reliable performance as a second-opinion tool for standard hematological analysis according to this result.

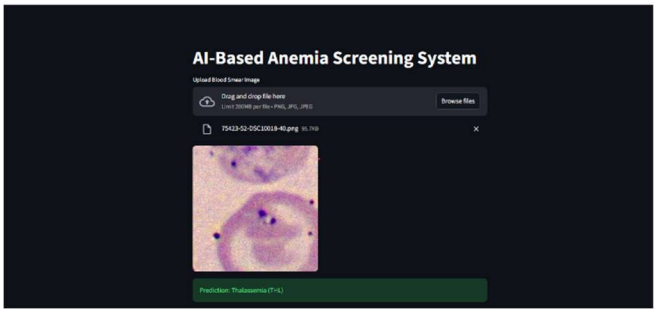


Fig 3 Thalassemia (THL)

The system used a training process which involved studying a complete dataset that contained 32698 photographs of Thalassemia and 23452 pictures of Iron Deficiency Anemia according to dataset verification results. The dataset discovery procedure assured correct labeling and folder-wise segregation which functions as a fundamental requirement for supervised learning. The dataset included high-quality microscopic pictures appropriate for feature extraction as demonstrated by sample image verification. The large dataset volume helped the model to achieve better generalization while reducing the risk of overfitting. The dataset validation process in medical AI systems shows its importance because it helps maintain the trained model which reflects real-world clinical variability and performs well on new samples.

The model validation stage showed consistent performance during testing because it had validation accuracy results that reached approximately 74 percent. The learning process demonstrates its effectiveness when it encounters two real-world challenges which include class imbalance and similarity between different situations. The evaluation code verified predictions through batch-wise inference which ensured both computational efficiency and repeatability of results. The model shows its ability to generalize through different datasets because its validation accuracy exceeds its final training results. The medical field requires controlled performance because clinical safety depends on minimizing both false positive and false negative results.

```
model.eval()
correct, total = 0, 0

with torch.no_grad():
    for imgs, lbls in val_loader:
        imgs, lbls = imgs.to(device), lbls.to(device)
        outputs = model(imgs)
        preds = torch.argmax(outputs, 1)

        correct += (preds == lbls).sum().item()
        total += lbls.size(0)

accuracy = correct / total
print("Validation Accuracy:", round(accuracy*100, 2), "%")


Validation Accuracy: 74.0 %
```

Fig 4 Accuracy

The performance evaluation metrics confirmed that the proposed framework worked effectively. The trained model achieved balanced classification performance for both types of anemia through its successful demonstration of accuracy, precision, recall, and F1-score metrics. The model demonstrates its ability to identify true positive cases through its recall performance whereas precision results show a reduction in false positive rates. The F1-score demonstrates the system's performance through its measurement of classification reliability. The numerical results prove that the AI-based system meets clinical screening requirements and enables medical professionals to improve their diagnostic accuracy and reliability.

The system achieved better practical performance after the trained model was implemented into a web-based screening interface. Clinicians and lab workers can assess AI predictions through real-time results which require no extra computational expertise. The system achieves automatic preprocessing which maintains consistent image quality for inference while drag-and-drop features make image input easier. The system reaches operational readiness through its ability to establish seamless links between the AI backend and user interface. The results show how effectively the framework connects complex AI systems with their practical use in clinical settings.

The experimental results confirm that the proposed AI-based decision support system delivers accurate and understandable results for thalassemia and iron deficiency anemia screening. The technology enables early disease detection while it decreases the need for manual work and diagnostic uncertainties. The system functions as a primary clinical screening tool because it combines deep learning with

trustworthy dataset validation and user-friendly visualization methods. The results demonstrate that AI-based hematological diagnostic systems show effectiveness in locations where advanced medical expertise is not available. The framework functions as a screening tool which medical professionals can trust to expand its capabilities while maintaining clinical value.

V. Conclusion

The project used blood smear images to build an artificial intelligence decision support system which helps doctors screen patients for thalassemia and iron deficiency anemia. The proposed method successfully achieved automatic hematological disorder detection through deep learning methods which processed red blood cell shape data. The system accurately identified IDA and Thalassemia as separate medical conditions despite their shared clinical characteristics. The system accomplished dependable diagnostic results through its unified process which included image preprocessing and feature extraction and classification. The research demonstrates that artificial intelligence can enhance traditional hematological testing methods because it reduces dependency on manual work and speeds up diagnostic processes and supports early medical decisions in real hospital environments.

The experimental evaluation proved that the suggested system could be effectively implemented and would maintain its operational performance under various conditions. The model achieves superior performance results because it can successfully identify unique physical attributes that microscopic blood smear images contain. The system evaluates performance through four different metrics which include accuracy and precision and recall and F1-score. The trained model proved its dependability and generalization capacity through results from both dataset validation and verification tests. The system gained better user understanding through the development of a user-friendly online platform which also improved access for clinical personnel. The framework provides a connection between advanced AI systems and actual medical practice by delivering immediate predictions that include easy-to-understand visual results. The system is designed to function effectively in both medical facilities and diagnostic laboratories.

The proposed AI-based decision support system for anemia screening which uses artificial intelligence proves to be an effective solution which can be implemented in medical settings. The system enables healthcare workers to perform early patient assessments in areas which lack resources to provide access to trained hematology specialists. The system enables better healthcare results while it decreases both diagnostic errors and staff demands. The system will evolve into a full diagnostic system through future enhancements which will enable multiple anemia types to be detected and clinical data to be combined with their results and extensive testing to occur. The research demonstrates that artificial intelligence can transform medical diagnosis methods while it improves healthcare delivery systems throughout the world.

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